

Using statistical learning models with household survey microdata

Applications to energy demand, energy poverty, and environmental taxation

Instructor: Manuel Tomás

Contact: manuel.tomas@bc3research.org

Date: Wednesday, June 10, 2026



EAERE Summer School in Resource and Environmental Economics 2026



EAERE

European Association
of Environmental and
Resource Economists

Recent applications of AI in Environmental
and Resource Economics



Session 1

Household surveys and main indicators

Why household surveys matter in environmental economics

They connect aggregate policy shocks with heterogeneous households

- **Policies affect households differently** because income, housing, location and fuels vary across the population.
- **Aggregate statistics** show total demand, prices and emissions, but **not who is vulnerable or exposed**.
- Survey **microdata covers multiple dimensions**: expenditure, income, dwelling characteristics, or demographics.
- For carbon pricing and environmental taxation, they allow **incidence simulations and compensation design**.

Key message

Microdata translate policy shocks into concrete population impacts: who pays, who is vulnerable, and who may need compensation.

The universe of EU household surveys

Different surveys answer different parts of the research question

Main public sources

Survey	Income	Consumption	Wealth	Labour
EU-SILC	Detailed	Missing	Partial	Partial
HBS / EPF	Partial	Detailed	Missing	Partial
HFCS	Partial	Partial	Detailed	Partial

- **EU-SILC (Eurostat)** focuses on income, housing conditions and deprivation indicators such as inability to keep the home warm or arrears on utility bills.
- **Household Budget Surveys (Eurostat)** measure detailed consumption and expenditure by category, which is why they are useful for energy-expenditure shares and tax-incidence exercises.
- **HFCS (ECB)** is richer on assets, liabilities and wealth, but less detailed on current consumption and domestic energy expenditure.

Key message

The best survey depends on the empirical object: consumption, living conditions, wealth or labour-market behaviour.

The Household Budget Survey (HBS)

Why HBS is central for this lecture

- The HBS record **household consumption expenditure** at a detailed product level.
- **Energy categories can be disaggregated** into electricity, gas and other domestic fuels; and transport fuel expenditure can also be studied separately.
- The same microdata support **distributional analysis** of energy prices, indirect taxes and first-round carbon-price burdens.
- **Limitations:** technical dwelling characteristics, actual physical consumption and subjective deprivation are often less complete than in energy-specific surveys.

Key message

HBS is strongest when the research question is about expenditure, energy demand, environmental taxation or heterogeneous household impacts of energy policies.

Typical research questions

The HBS can support demand modelling, vulnerability analysis and policy simulation

Energy demand

- Which household and dwelling characteristics explain electricity or gas spending?
- How does demand vary by climate zone, income, household size or heating technology?
- Can we predict household-level energy expenditure from observable variables?

Energy poverty

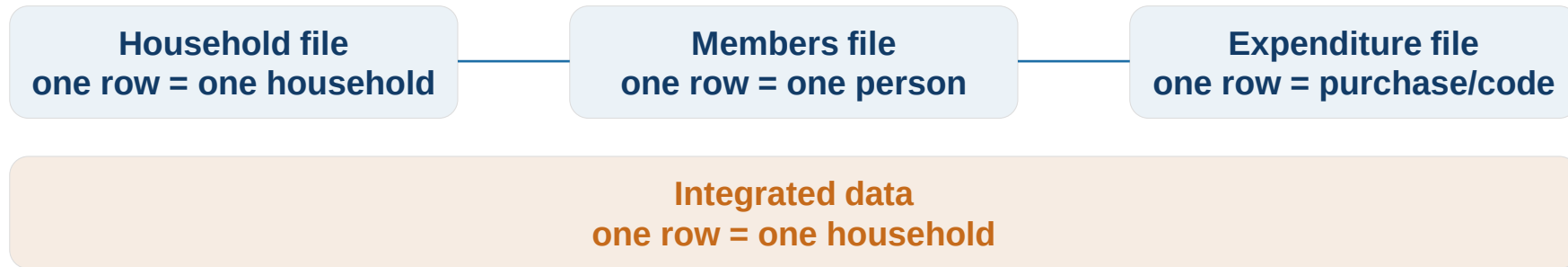
- Which households face high energy burdens or report inability to keep the home warm?
- Can observable characteristics predict vulnerability?
- How do indicators change when thresholds or equivalence scales change?

Environmental taxation

- Which households are more exposed to energy-price or carbon-tax increases?
- How progressive or regressive are first-round burdens?
- Which recycling scheme best protects vulnerable groups?

Anatomy of household consumption microdata

What each row and variable usually represents



- The main **unit of observation is the household**, but household, expenditure and person-level files need to be merged carefully.
- **Core variables** usually include income, expenditure categories, region, household composition, age and education of the reference person, tenure and housing characteristics.
- **Energy variables** may appear as expenditure by fuel, physical quantities, subjective deprivation, energy-efficiency characteristics or appliance ownership.

Sample weights and representativeness

Descriptive statistics and model evaluation are related but not identical operations

Key idea

The HBS is a complex survey: households have different sampling weights.

Each observation represents a different number of households in the population.

- Survey weights make **sample estimates representative of the target population**. They are essential for descriptive rates, totals and distributional indicators.
- For modelling, weights require a substantive decision: **some algorithms accept case weights directly, others do not**.
- In the classroom examples, **we keep models unweighted** and then use weights for population-level summaries and policy interpretation.

Example

Weighted mean: $\bar{y}_w = \sum_i w_i y_i / \sum_i w_i$

Computing relevant indicators

Weighted indicators translate household-level measures into population-representative policy evidence

Key formulas

Energy burden: $EB_i = E_i / Y_i$

Energy-poor rate: $EP = \sum_i w_i \cdot 1[EB_i > \tau \text{ and } Y_i < z] / \sum_i w_i$

Group impact: $Impact_g = \sum_{i \in g} w_i \cdot \Delta C_i / \sum_{i \in g} w_i$

- **E_i and Y_i** are household domestic energy expenditure and disposable income.
- **τ and z** define the high-burden and low-income thresholds.
- **w_i and ΔC_i** are the survey weight and the extra cost from a price or tax shock.

Key message

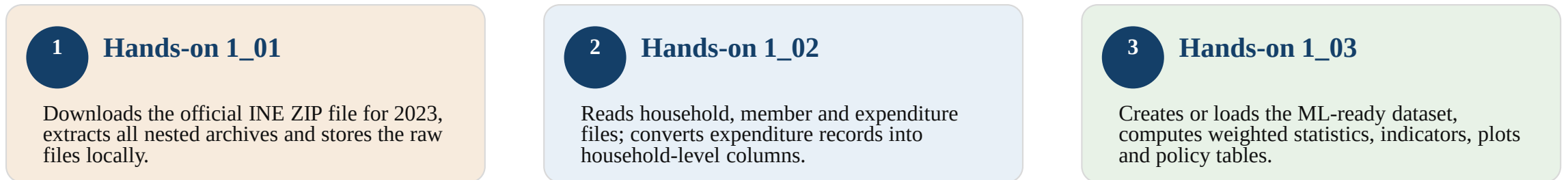
Weights convert household indicators into population-representative rates, averages and distributional impacts.

Hands-on 1: Starting with household consumption microdata of Spain

From official HBS microdata to policy indicators and ML-ready variables



Brief description of the scripts



Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_01 — download and organise the HBS raw files

Set project

Use `here()` to locate the project root and read `vectors.csv` for auxiliary names.

Choose year

In the course version, `initial.year = 2023` and `final.year = 2023`.

Download ZIP

Build the INE URL: `base.url + datos_2023.zip` and save it under `data/2023`.

Extract

Use `archive::archive_extract()` for the main and nested ZIP files.

Hands-on 1_02 — from raw survey files to household tables

Raw files

`hbs_h`
`hbs_m`
`hbs_e`

Read

Separator changes
by year

Reshape

Expenditure rows
→ columns

Save

`hbs_h_2023`
`hbs_m_2023`

Key transformation

Each ECOICOP expenditure code becomes a household-level column

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Variable group	Examples	Role
Identifiers	survey_year, household_id	metadata / ID
Weight	survey_weight = FACTOR	population summaries
Total budget	total_consumption_spend_eur	control
Domestic energy	home_energy_spend_eur; electricity, gas, other fuels	targets / features
Vehicle fuels	private_vehicle_fuel_spend_eur; diesel, petrol	target / features
Income	monthly_net_income_eur	predictor

Key transformation

This selection keeps the course dataset compact while preserving the variables needed for modelling energy demand and poverty.

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Data dictionary I

ML variable	HBS source	Meaning	Role
survey_year	ANOENC	Survey year.	metadata
household_id	NUMERO	Anonymous household identifier.	ID
survey_weight	FACTOR	Population expansion factor; use for representative estimates.	weight
total_consumption_spend_eur	EUR_HE00	Total annual household consumption expenditure in euros.	control
monthly_net_income_eur	IMPEXAC	Monthly net regular household income in euros.	predictor

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Data dictionary II

ML variable	HBS source	Meaning	Role
home_energy_spend_eur	EUR_HE04511–EUR_HE04549	Annual domestic energy expenditure: electricity, gas, liquid fuels and solid fuels.	target
home_electricity_spend_eur	EUR_HE04511, EUR_HE04512	Annual electricity expenditure for the main dwelling and other dwellings.	feature
home_gas_spend_eur	EUR_HE04521–EUR_HE04524	Annual expenditure on natural/city gas and LPG.	feature
home_other_fuels_spend_eur	EUR_HE04531/32, EUR_HE04541/42, EUR_HE04548/49	Annual expenditure on liquid fuels, coal and other solid fuels for home use.	feature
private_vehicle_fuel_spend_eur	EUR_HE07221–EUR_HE07223	Annual private vehicle fuel expenditure: diesel, petrol/gasoline and other vehicle energy.	target

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Data dictionary III

ML variable	HBS source	Meaning	Role
vehicle_diesel_spend_eur	EUR_HE07221	Annual expenditure on diesel for private vehicles.	feature
vehicle_petrol_spend_eur	EUR_HE07222	Annual expenditure on petrol/gasoline for private vehicles.	feature
household_size	NMIEMB	Number of household members.	predictor
children_under18	NMIEM5	Number of household members under 18 years old.	predictor
employed_members	NUMOCU	Number of employed household members.	predictor

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Data dictionary IV

ML variable	HBS source	Meaning	Role
region_code	CCAA	Autonomous Community of residence; useful as a geography/climate proxy.	predictor
municipality_size_code	TAMAMU	Municipality size category; useful as an urban/rural proxy.	predictor
dwelling_area_m2	SUPERF	Usable dwelling area in square metres; code -9 should be set to missing.	predictor
heating_energy_source	CALEF + FUENCALE	Readable label for the main heating energy source, including “No heating”.	predictor
hot_water_energy_source	AGUACALI + FUENAGUA	Readable label for the main hot-water energy source, including “No hot water”.	predictor

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Computing relevant indicators

Key formulas

Energy burden: $EB_i = E_i / Y_i$

10% poverty rate: $EP_{10} = \sum_i w_i \cdot 1[EB_i > 0.10] / \sum_i w_i$

Group impact: $Impact_g = \sum_{i \in g} w_i \cdot \Delta C_i / \sum_{i \in g} w_i$

- **E_i and Y_i** are annual domestic energy expenditure and annual net household income.
- **w_i** is the HBS sampling weight used for representative estimates.
- **ΔC_i** is the extra cost from a policy or price shock, such as a gas-price increase.

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Computed indicators

Indicator	Script variable	Policy question
Energy burden	home_energy_burden	How large is energy spending relative to income?
10% energy poverty	energy_poor_10pct	Who spends more than 10% of income on domestic energy?
Gas shock cost	gas_price_shock_extra_cost_eur	How much does a +30% gas-price increase cost?
Affordability gap	affordability_gap_10pct_eur	How many euros would bring a household below the 10% rule?
Transport fuel burden	private_transport_fuel_burden	Who is exposed to private vehicle fuel costs?

Extension *The same structure can be reused for other household-level burdens: electricity, gas, transport fuels or carbon-tax exposure.*

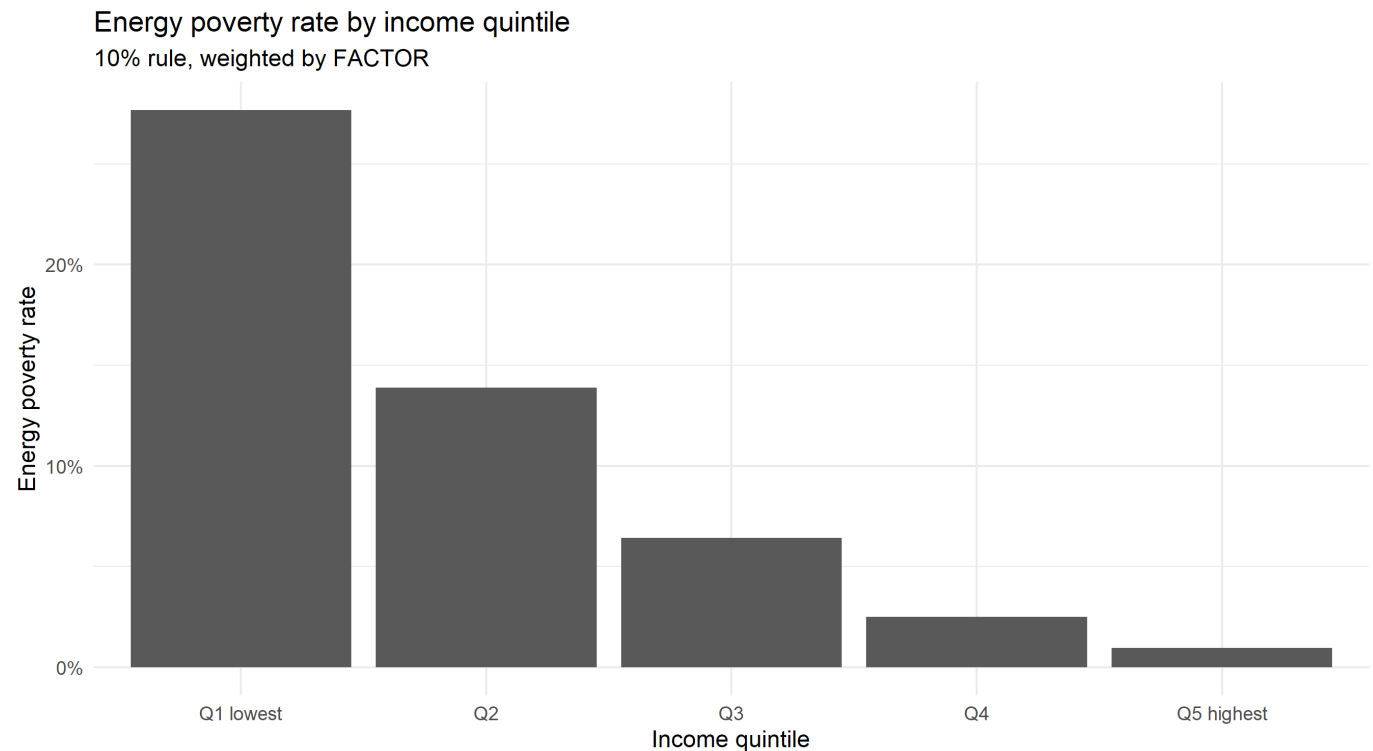
Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Energy burden and the 10% energy poverty indicator

```
home_energy_burden = home_energy_spend_eur  
/ annual_net_income_eur  
energy_poor_10pct = 1 if  
home_energy_burden > 0.10
```

Quintile	Burden	Poor
Q1 lowest	8.6%	27.7%
Q2	6.1%	13.9%
Q3	4.7%	6.4%
Q4	3.9%	2.5%
Q5 highest	3.0%	1.0%



Result: the 10% indicator is strongly concentrated in the lowest income quintile.

Hands-on 1: Starting with household consumption microdata of Spain

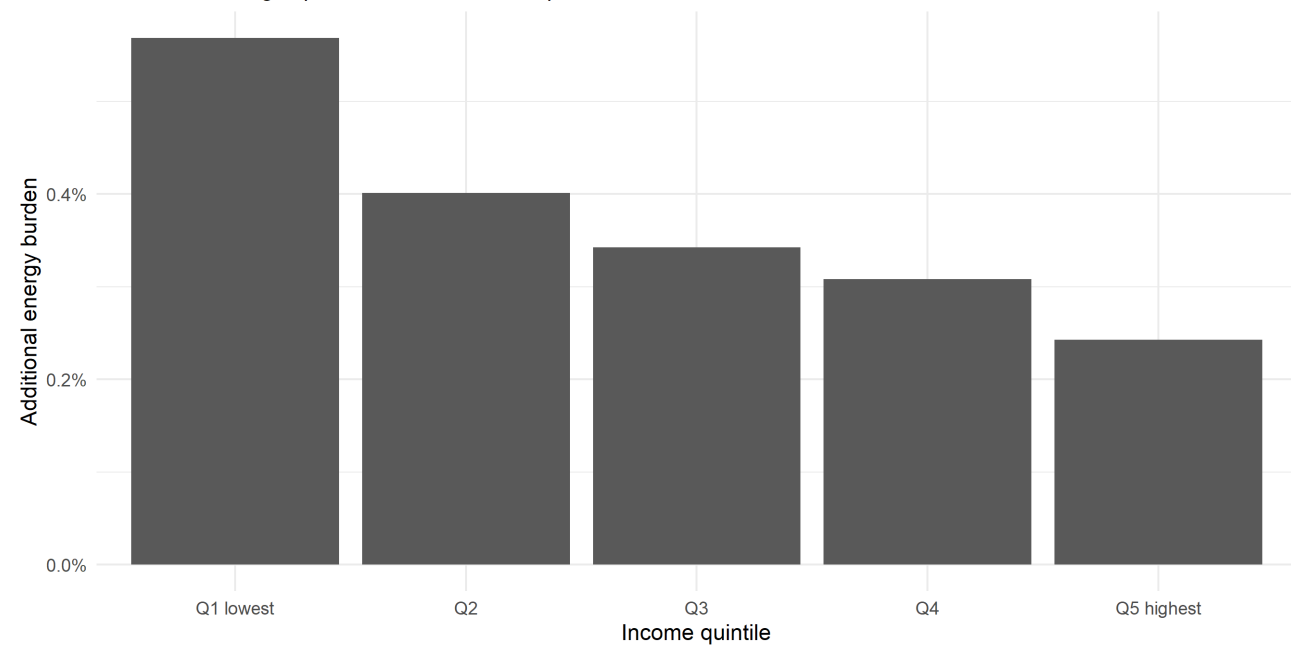
Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Distributional impact of a gas price increase

```
extra_cost = home_gas_spend_eur × 0.30  
new_burden = (home_energy_spend_eur +  
extra_cost) / annual_income
```

Q	Extra cost	Burden
Q1	€60	+0.57 pp
Q2	€73	+0.40 pp
Q3	€92	+0.34 pp
Q4	€109	+0.31 pp
Q5	€133	+0.24 pp

Distributional impact of a gas price increase
Scenario: +30% gas prices, no household response



Absolute extra cost rises with income, but the burden impact is largest for low-income households because income is the denominator.

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

Exercise: build one more policy indicator

Private transport fuel burden shows how the same logic can be reused for a new policy question

Transport fuel burden: $TFB_i = \text{private_vehicle_fuel_spend}_i / \text{annual_income}_i$

- Which income quintile has the highest rate of high private transport fuel burden?
- Is the pattern the same as for domestic energy poverty?
- What changes if the threshold is 3% instead of 5%?

Hands-on 1: Starting with household consumption microdata of Spain

Hands-on 1_03 — creates hbs_energy_ml_2023.csv

One solution

```
transport_fuel_burden = private_vehicle_fuel_spend_eur /  
annual_net_income_eur  
high_transport_fuel_burden_5pct = 1 if burden > 0.05
```

High transport fuel burden rate

